



Zhongzhi Chemical Technology

众智化工科技 · Inorganic Silicon



Shandong Zhongzhi Chemical Technology Co., Ltd.

Unity for Progress, Wisdom for New Horizons

众新聚力 · 智拓新局

Sodium Silicate & Metasilicate · Export Product Brochure 2026

100,000+ T

Annual Capacity

30,000 m²

Factory Area

71+

National Patents

ISO / REACH

Certified



01

Company Profile

Inorganic silicon compounds specialist since 2011

Shandong Zhongzhi Chemical Technology Co., Ltd. (Zhongzhi Tech) was founded in 2011 in Changyi, Shandong. We focus on **inorganic silicon compounds** — sodium silicates, metasilicates, potassium / lithium silicates and silicon-based flame retardants. Zhongzhi is a national-level "Little Giant" enterprise, a national high-tech enterprise, and vice-chairman unit of the China Inorganic Silicon Compounds Association.

100,000+ T

Annual Capacity

30,000 m²

Smart Factory

71+ Patents

R&D Strength

15 Platforms

Innovation Hubs

- Main drafter of industry standard HG/T 2568-2021 Industrial Sodium Metasilicate
- Silicon-based flame retardant: National Manufacturing Single Champion Product
- Certifications: ISO 9001:2015 · ISO 14001 · REACH Registered
- Export port: Qingdao · MOQ from 1 MT · Lead time ~2 weeks · 30+ countries served



02

Product Portfolio

Nine core product lines for global partners

01 Anhydrous Sodium Metasilicate $\text{Na}_2\text{O} \geq 51\%$ · STPP alternative	02 Sodium Metasilicate Pentahydrate Na_2O 28.5–30% · Crystalline	03 Instant Powder Sodium Silicate LGS-20 / LGS-30 · Fast dissolve
04 Liquid Sodium Silicate 6 grades · Modulus 1.3–3.4	05 Potassium Silicate 3 grades · Low S/P electronic	06 Lithium Silicate Concrete densifier & hardener
07 Sodium Orthosilicate $\text{Na}_2\text{O} \geq 58.5\%$ · Strongest alkali	08 Sodium Silicate (Water Glass) Solid & liquid · Custom modulus	09 Silicon-Based Flame Retardant LOI $\geq 29\%$ · Single Champion



03

Anhydrous Sodium Metasilicate



Anhydrous Sodium Metasilicate

High-purity $\text{Na}_2\text{O} \geq 51\%$. White granular / powder. Total alkali + silica > 95%. Excellent detergency, pH buffering and emulsifying. Effective STPP replacement; compatible with peroxides and bleaching agents. Molecular formula Na_2SiO_3 , MW 122.06, m.p. 1089 °C.

Detergent

Metal Cleaning

Industrial Cleaning

Construction

Ceramics

Textile

Oil Drilling

Paper

SPEC	Granular		Powder
	Superior	Normal	
$\text{Na}_2\text{O} \%$	51.00 ± 1.00	51.00 ± 1.00	50.50 ± 1.00
$\text{SiO}_2 \%$ $\geq$	45.00	45.00	45.50 ± 1.00
$\text{Fe} \%$ $\leq$	0.0150	0.0200	0.0150
Bulk Density g/cm^3	1.00–1.35	1.00–1.35	1.00–1.35
Particle Size	18–60 mesh $\geq 93\%$	18–60 mesh $\geq 93\%$	>60 mesh $\geq 85\%$
Whiteness $\geq$	85	80	90
Water Insoluble $\%$ $\leq$	0.15	0.20	—
Transparency $\%$ $\geq$	80	60	—

Key Properties

- $\text{Na}_2\text{O} \geq 51\%$ with total alkali + silica exceeding 95%
- No water of crystallization — ideal for powder detergents
- Effective phosphate-free STPP replacement for eco-friendly formulas
- Strong free alkalinity and hard-water softening (Ca^{2+} / Mg^{2+} sequestration)

MOQ 1 MT	Lead Time ~2 Weeks	Port Qingdao	OEM Available	Sample Free
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Packing: 25 kg PP / kraft bag · 500 kg / 1000 kg FIBC jumbo · OEM labels available



04

Sodium Metasilicate Pentahydrate



Sodium Metasilicate Pentahydrate

Na₂O 28.5–30%. White crystalline solid, m.p. 72.2 °C, MW 212. Strong detergency, pH buffering, emulsifying. Corrosion protection for Al/Zn. Bulk density 0.8–1.0 g/cm³. Widely used in detergents, metal cleaning, ceramics and oilfield mud.

Detergent

Metal Cleaning

Construction

Oil & Gas

Paper

Ceramics

Textile

Food Equipment

SPEC	Granular		Powder	
	Superior	Normal	Superior	Normal
Na ₂ O %	28.70–30.00	28.50–30.00	28.70–30.00	28.50–30.00
SiO ₂ %	27.80–29.20	27.50–29.20	27.80–29.20	27.50–29.20
Fe % ≤	0.0090	0.0150	0.0150	0.0150
Bulk Density g/cm ³	0.8–1.0	0.8–1.0	0.8–1.0	0.8–1.0
Particle Size	14–60 mesh ≥95%	14–60 mesh ≥95%	>60 mesh ≥15%	>60 mesh ≥15%
Whiteness ≥	80	75	85	80
Water Insoluble % ≤	0.05	0.10	0.10	0.10
Transparency % ≥	85	75	85	75

Key Properties & Applications

- Crystalline form — easy handling, good free-flow for automatic dosing
- Buffer alkalinity protects Al / Zn / glass / porcelain in wash liquor
- Ceramic slip deflocculant — improves fluidity and casting quality
- Also used in acid-resistant mortar, paper deinking, textile pretreatment

MOQ 1 MT	Lead Time ~2 Weeks	Port Qingdao	OEM Available	Sample Free
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Packing: 25 kg bags · jumbo bags · OEM available · Port Qingdao

05

Instant Powder Sodium Silicate



Instant Powder Sodium Silicate

LGS-20 & LGS-30. White free-flowing powder with fast cold-water dissolution. Ideal for automated batching, high-concentration liquids and refractories. Excellent cold resistance and uniform blending with other powders.

Detergent

Metallurgy

Refractories

Construction

Ceramics

Oil Drilling

Soap

Power Gen

SPEC	LGS-20		LGS-30	
	Superior	Normal	Superior	Normal
(Na ₂ O+SiO ₂) % ≥	77.00	76.00	80.00	79.00
Bulk Density g/cm ³ ≥	0.45	0.30	0.60	0.45
Whiteness ≥	90	80	90	80
Soluble Rate (60 °C) ≤	60 s	100 s	—	—
Fe % ≤	0.015	—	0.015	—
Water Insoluble % ≤	0.05	0.10	0.10	0.10
Modulus (M.R.)	1.90–2.10	1.90–2.10	2.90–3.10	2.90–3.10

Why Instant Powder

- Rapid dissolution at ambient / mild temperature — no long cooking
- Suitable for mechanical and automated dosing systems
- Blends uniformly with other dry ingredients for high-conc. liquids
- Used as binder / accelerator in refractories, precision casting, drilling

MOQ 1 MT	Lead Time ~2 Weeks	Port Qingdao	OEM Available	Sample Free
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Packing: 25 kg bags · jumbo bags · dry cool storage · Shelf life 24 months



06

Liquid Sodium Silicate (Water Glass)



Liquid Sodium Silicate (Water Glass)

Six grades, modulus 1.3–3.4. Clear viscous aqueous solution $\text{Na}_2\text{O} \cdot n\text{SiO}_2$. Produced by dissolving silicate glass under heat and pressure. Strong adhesion, film-forming and adjustable chemistry for construction, automotive, aquaculture, textiles and wastewater treatment.

Construction

Water Treatment

Detergent

Metal Protection

Textile

Aquaculture

Wood

Food

SPEC	Low Modulus			High Modulus		
	LGY33	LGY501	LGY502	LGY401	LGY402	LGY403
Baumé (20 °C)	32.5–35.0	50.0–51.0	50.0–51.0	39.0–40.0	39.0–40.0	39.0–40.0
$\text{Na}_2\text{O} \% \geq$	10.50	14.20	12.80	8.30	8.30	8.20
$\text{SiO}_2 \% \geq$	14.50	28.00	29.40	26.50	26.50	26.00
Water Insoluble % $\leq$	0.10	0.20	0.20	0.20	0.80	—
Modulus (M.R.)	1.30–1.50	1.90–2.20	2.20–2.50	3.20–3.40	3.20–3.40	3.20–3.40

Grade Guide & Applications

- LGY401: lowest insolubles — critical / premium applications
- LGY402: industrial general purpose · LGY403: cost-sensitive projects
- LGY501 / LGY502: high Baumé low-modulus for heavy industrial use
- Uses: concrete set / strengthen · metal protective film · wastewater coagulant · wood FR · dye assist

MOQ 1 MT	Lead Time ~2 Weeks	Port Qingdao	OEM Available	Sample Free
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Packing: 200 L drum · 1000 L IBC · 25 MT ISO tank · custom packing available



07

Potassium Silicate



Potassium Silicate

Three grades, modulus 3.3–4.9. Premium aqueous solution with excellent adhesion and higher thermal stability than Na-silicates. Low S/P for electronic grade. Film-forming and chemical-resistant — preferred binder for refractories and welding electrodes.

Refractory

Welding Electrodes

Oilfield Drilling

Coatings

Flame Retardants

Displays

Vat Dyes

Pharma

SPEC	Potassium Silicate		
	Grade A	Grade B	Grade C
SiO ₂ %	22.0–23.0	19.0–19.8	16.0–16.8
K ₂ O %	9.8–10.9	7.4–8.2	5.1–5.6
Solid Content %	31.8–33.9	26.4–28.0	21.1–22.4
Density g/cm ³	1.28–1.33	1.23–1.26	1.16–1.18
pH Value	12.9–13.2	12.0–12.3	11.6–12.0
Sulfur % ≤	0.01	0.01	0.01
Phosphorus % ≤	0.01	0.01	0.01
Water Insoluble % ≤	0.05	0.05	0.05
Modulus	3.3–3.5	3.8–4.0	4.7–4.9

Key Properties

- Higher thermal stability than sodium silicates
- Low sulfur / phosphorus — suitable for electronic-grade uses
- Excellent adhesion for refractory bricks and welding rod coating
- Also used in inorganic coatings, fireproof agents, fluorescent screen layers

MOQ 1 MT	Lead Time ~2 Weeks	Port Qingdao	OEM Available	Sample Free
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Packing: drums / IBC · OEM available · FOB / CIF Qingdao



08

Lithium Silicate



Lithium Silicate

$\text{Li}_2\text{O} \cdot n\text{SiO}_2$. Colorless to slightly yellow liquid. Low molecular weight enables deep penetration; reacts with free Ca^{2+} in concrete to form stable C-S-H gel. UV resistant, does not yellow. Enhances hardness, abrasion resistance and impermeability.

Concrete Densifier

Floor Hardeners

Coatings

Building Protection

Refractory

Casting

Ceramics

Electronics

SPEC	Lithium Silicate
Chemical Name	Lithium Silicate (Lithium Water Glass)
Molecular Formula	$\text{Li}_2\text{O} \cdot n\text{SiO}_2$
Appearance	Colorless to slightly yellow transparent liquid
Form	Aqueous solution
Key Performance	Deep penetration · Fast reaction · Uniform C-S-H film
Surface Benefits	Hardness ↑ · Abrasion resistance ↑ · UV stability
Main Use	Concrete densifier · Floor hardener · Building protection

Why Lithium vs Sodium / Potassium

- Deeper penetration and more uniform densification
- Superior UV and weather resistance — no yellowing
- Reduces dusting, chalking and surface cracking
- Ideal for industrial floors, commercial flooring and façade protection

MOQ 1 MT	Lead Time ~2 Weeks	Port Qingdao	OEM Available	Sample Free
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Packing: 200 L drum · 1000 L IBC · solid form on request



09

Sodium Orthosilicate



Sodium Orthosilicate

$\text{Na}_2\text{O} \geq 58.5\%$ — strongest alkaline silicate in our range. White granular / powder. Non-toxic, odorless. Superior detergency, emulsifying and pH buffering for heavy-duty industrial cleaning, metal treatment and ceramic grinding.

Industrial Cleaning

Metal Treatment

Ceramic Grinding

Refractory

Textile

Soil Stabilization

Paper

Soap

SPEC	$\text{Na}_2\text{O} \% \geq$	$\text{SiO}_2 \% \geq$	$\text{Fe} \% \leq$	Whiteness $\geq$	Bulk Density $\geq$
Value	58.50	27.50	0.010	80.00	0.90 g/cm ³

Key Properties & Uses

- Highest free alkalinity among common sodium silicates
- Heavy-duty degreasing and metal surface preparation
- Ceramic grinding aid and refractory raw material
- Also used in soil stabilization, paper deinking and soap filler

MOQ 1 MT	Lead Time ~2 Weeks	Port Qingdao	OEM Available	Sample Free
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Packing: 25 kg bags · jumbo bags · OEM · Port Qingdao · Lead time ~2 weeks



10

Sodium Silicate (Water Glass)



Sodium Silicate (Water Glass)

Liquid & solid forms produced from silica sand + soda ash. The most versatile industrial silicate — adhesive, binder, deflocculant, detergent builder. Serves 20+ industries. Modulus adjustable 1.6–3.5 on request.

Detergents

Construction

Paper

Foundry

Water Treatment

Welding

Textile

Ceramics

SPEC	Solid		Liquid	
	Standard	Custom	Standard	Custom
Na ₂ O %	19.0–28.0	Adjustable	7.0–14.0	Adjustable
SiO ₂ %	57.0–66.0	Adjustable	22.0–33.0	Adjustable
Modulus (M.R.)	2.0–3.3	1.6–3.5	2.2–3.3	1.6–3.5
Fe % ≤	0.02	—	—	—

Applications Snapshot

- Detergent & soap builder · water softener · settling aid
- Foundry sand binder · grinding wheel · metal anti-corrosion
- Fast-set cement · acid-resistant cement · soil stabilizer · refractory
- Paper, textile dyeing, mining, waterproofing and catalyst carriers

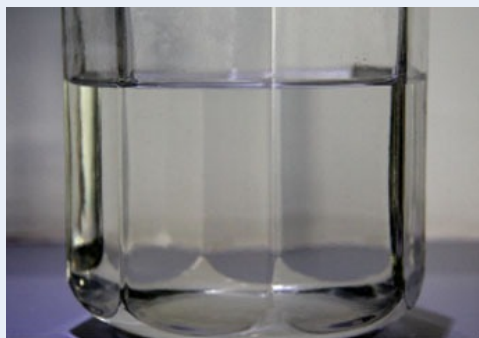
MOQ 1 MT	Lead Time ~2 Weeks	Port Qingdao	OEM Available	Sample Free
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Packing: bags / drums / IBC / ISO tank · custom modulus & packing available



11

Silicon-Based Flame Retardant



Silicon-Based Flame Retardant

LOI $\geq$ 29%. New-generation water-soluble, non-toxic silicon flame retardant. Anti-dripping, smoke-suppressing. National Manufacturing Single Champion Product. Liquid & powder forms for textiles, plastics, cables and fireproof materials.

Textile FR

Polymer

Cables

Engineering Plastics

Insulation

Coatings

Electronics

Automotive

SPEC	Novel Silicon-Based	Powder Form	
		Premium	Grade I
Appearance	Colorless to light yellow	White solid powder	White solid powder
SiO ₂ % $\geq$	15.0	15.0	15.0
Heavy Metals & PBBs	$\leq$ 5.0 mg/kg	$\leq$ 5.0 mg/kg	$\leq$ 5.0 mg/kg
LOI (Textile) $\geq$	29%	29%	29%

Key Advantages

- High LOI $\geq$ 29% for reliable textile and polymer FR performance
- Anti-dripping — prevents molten droplet spread in fire
- Smoke suppression; halogen-free, non-toxic, water-soluble
- Improves processability, heat resistance and mechanical properties

MOQ 1 MT	Lead Time ~2 Weeks	Port Qingdao	OEM Available	Sample Free
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Packing: 25 kg bags · jumbo bags · liquid packaging on request · Port Qingdao



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Contact & Export Service

Shandong Zhongzhi Chemical Technology Co., Ltd.

Changyi City, Shandong Province, China

Website: www.zzhchem.com | Email: info@zzhchem.com

Port: Qingdao, China | MOQ: 1 MT | Lead Time: ~2 Weeks

ISO 9001:2015 · ISO 14001 · REACH Registered

Quotation · SDS · TDS · COA · Sample → www.zzhchem.com/contact

Packing Options

- 25 kg PE / PP bags
- 200 L / 250 kg drums
- ISO tank containers
- 500–1000 kg jumbo bags
- 1000 L IBC totes
- Custom OEM labels



Zhongzhi intelligent manufacturing base — Changyi, Shandong, China

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